



Osnias-ID: A Unique Participant Identifier for Multi-Chain Clearing

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OSNIAS CLEARING**Osnias-ID:****A Unique Participant Identifier for Multi-Chain Clearing***Clearing Participant Identifier · Technical White Paper & Public Specification*

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Abstract

Osnias-ID is a project-specific, chain-independent identifier for clearing accounts within the Osnias Clearing network. It is designed to provide a compact and persistent reference that can be used across multiple blockchain environments without depending on any specific wallet address or chain-native account format.

Each Osnias-ID is linked through the Osnias Registry to an identified participant, such as a company, institution or individual, and to one or more authorized technical addresses. A single legal entity may therefore operate several Osnias-ID clearing accounts while retaining the same underlying legal identity.

The architecture separates three layers: legal identity, clearing-account identification and technical execution addresses. This separation supports registry lookup, access control, counterparty resolution, auditability and interoperability, while reducing the amount of identity and addressing data that must be transported on-chain or across the network. Osnias-ID is not a legal-entity identifier, bank account number, wallet address, regulatory authorization number, security or token.

Keywords

Osnias Clearing, Osnias-ID, Clearing, Clearing Participant Identifier, Blockchain, On-chain Clearing, Multilateral Netting, Digital Identity, Participant Registry, Wallet Identity, KYC, KYB, AML, Financial Infrastructure, Smart Contracts, Interoperability, ISO 20022, MOD-97 Checksum, Institutional Blockchain

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1. Original Design Intent: Why Osnias-ID Is Needed

Osnias-ID is initially designed to provide Osnias Clearing with a compact, chain-independent participant identifier that can be used consistently across a multi-chain clearing environment.

Its original objectives are the following:

1- Provide a chain-independent identifier

Osnias-ID is designed as an autonomous identifier format that does not depend on the addressing system, account model or technical conventions of any specific blockchain.

A participant therefore keeps the same Osnias-ID even if the participant operates across several supported networks or changes an authorized technical address.

2- Reduce on-chain and network resource consumption

Instead of transporting complete identity records, multiple wallet addresses or chain-specific metadata through each clearing message, the protocol may transport only the 16-digit Osnias-ID where appropriate.

The detailed participant information is resolved through the Osnias registry. This approach is intended to reduce message size, on-chain data consumption and unnecessary duplication of identity data across the network.

3- Identify each participant independently of the underlying blockchain

Osnias-ID provides a persistent participant reference above the blockchain layer. The same user or institution may therefore be identified consistently regardless of whether the underlying technical interaction occurs through Ethereum, Sei EVM, Arbitrum, Avalanche, Base, BNB Smart Chain, Solana, TRON, Stellar/Soroban or another network subsequently approved by Osnias Clearing.

4. Provide a participant directory for clearing counterparties

Osnias-ID is associated with a registry that allows authorized users and system components to resolve and identify participants interacting with the clearing platform.

In particular, when a participant issues a receivable, invoice or clearing request, the counterparty can identify the issuer through the Osnias-ID rather than relying solely on a blockchain wallet address.

The registry therefore provides an operational identity layer for participants creating, receiving or settling claims within Osnias Clearing.

2. Purpose and Scope

The purpose of Osnias-ID is to identify a clearing participant within the Osnias network independently of the technical address used to send, receive or settle a clearing instruction. This separation allows a participant identity to remain stable while addresses, chains, wallets or operational keys change under controlled procedures.

- Participant identification and registry lookup.
- Binding between an Osnias-ID and authorized wallet address(es).
- Admission control for payment and clearing requests.
- Counterparty resolution before execution.
- Audit trail for identity-to-address changes.
- Optional interoperability metadata for external identifiers such as LEI or jurisdiction-specific business identifiers.

3. Status of the Specification

This document is a public preprint / working specification. It defines the Osnias-ID data model and a deterministic checksum rule for implementation and independent testing. It does not claim external certification or regulatory approval.

VERSION STATUS

v1.0 is suitable for public technical archiving and preprint publication. The checksum rule is now normative within this document; external conformity claims remain explicitly excluded.

4. Identifier Structure

The canonical human-readable form proposed is: **KK-NNN-AAAAAAAA-XX**

Field	Length	Authority	Meaning
KK	2 digits	Osnias	Osnias-assigned blockchain/network code.
NNN	3 digits	Osnias	Osnias-assigned server participant or institutional allocation code.
AAAAAAAA	9 digits	Server	Server-assigned sequential or controlled unique participant value.
XX	2 digits	Server	Server-calculated checksum according to §14.

5. Identity and Addressing Are Separate Layers

An Osnias-ID identifies a participant record. A wallet address identifies a technical destination or signing account. They must not be treated as equivalent identifiers.

- Osnias-ID: stable clearing participant reference.
- Wallet address: technical execution endpoint.
- KYC/KYB record: identity evidence and verification data.
- External identifiers: optional interoperability metadata.

A participant may change an operational wallet without changing its Osnias-ID, provided the registry change follows the applicable verification and audit procedure.

6. Registry Architecture

The Osnias registry is the authoritative resolution layer between Osnias-ID and the participant's current authorized operational data. The registry may store private and public fields in separate data domains.

- Canonical Osnias-ID.
- Current authorized wallet address or addresses.
- Participant status: active, suspended, revoked or pending.
- Identity verification status and timestamp.
- Jurisdiction and legal-name fields where required.
- Optional LEI, UID or other external identifier metadata.
- Creation and update audit timestamps.
- Historical wallet bindings and revocation records.

PRIVACY BOUNDARY

Personal or confidential KYC/KYB evidence should not be written to a public blockchain merely because the Osnias-ID itself is used in an on-chain or message-layer workflow.

7. Access Control: Osnias-ID required for Clearing Participants

A clearing request must not be admitted solely because a wallet is connected. Before the request is accepted, the system must resolve the initiating and receiving participant through the registry and verify the relevant Osnias-ID status.

1. Resolve the wallet address to an Osnias-ID.
2. Verify that the Osnias-ID exists and is active.
3. Verify that the wallet is currently authorized for that Osnias-ID.
4. Apply any network, jurisdiction, lender, escrow or transaction-specific eligibility rules.
5. Only then admit the clearing request to the message or settlement workflow.

CLEARING ADMISSION INVARIANT

If the initiating or required counterparty Osnias-ID cannot be resolved and validated, the clearing request must fail closed.

8. One Legal Entity, Multiple Osnias-ID Clearing Accounts

An Osnias-ID identifies a **clearing account within the Osnias Clearing network**. It is not a legal-entity identifier.

A company or institution may therefore hold **multiple Osnias-ID clearing accounts**, while all of them remain linked in the Osnias Registry to the same legal entity.

The legal entity may be identified through an external identifier such as a national company registration number, a French SIREN, a Swiss UID/CHE, a LEI, or another recognized jurisdiction-specific identifier.

The architecture therefore separates three levels:

- **Legal identifier** → identifies the company or legal person.
- **Osnias-ID** → identifies a clearing account.
- **Wallet address** → identifies the technical execution endpoint.

A single legal entity may therefore operate several Osnias-ID accounts for different clearing activities, currencies, cycles, mandates or technical environments.

Core principle:

An Osnias-ID is not a company identifier. It is the identifier of a clearing account linked to an identified participant.

9. Cross-Chain Use

Osnias-ID is chain-independent as an identity reference. The wallet-address layer may contain chain-specific endpoints. An Osnias-ID therefore provides a stable clearing identity even where the technical architecture uses multiple EVM or non-EVM environments.

The identifier itself does not grant cross-chain rights. Each chain-specific action remains subject to the protocol rules, contract permissions and registry state applicable to that environment.

10. External Identifier Interoperability

Osnias-ID may reference external identifiers as metadata when useful for institutional resolution. Such linkage does not replace the external identifier and must never be presented as an official allocation by the external authority.

External reference	Possible use	Boundary
LEI	Legal-entity cross-reference	Osnias does not issue LEIs.
Swiss UID / CHE	Swiss enterprise cross-reference	Osnias does not issue UID numbers.
FINMA register data	Regulatory-status metadata where publicly applicable	No FINMA endorsement or authorization is implied.
Other national identifiers	Jurisdiction-specific matching	Subject to source and privacy constraints.

11. Relationship to ISO 20022

Osnias-ID is not an ISO 20022 identifier and is not claimed to be an ISO 20022-certified data element. The project may use ISO 20022 concepts and structured-data practices as interoperability references when mapping participant, account, agent or payment-message data.

At the date of this document, ISO 20022-1:2026 and ISO 20022-3:2026 are published editions of the ISO 20022 series. Their existence does not confer standard status on Osnias-ID.

12. Allocation hierarchy

Allocation authority is intentionally split so that not every field is generated by the same component.

Level	Field	Authority	Control objective
1	KK	Osnias Clearing	Allocate and maintain blockchain/network

			codes.
2	NNN	Osnias Clearing	Allocate participant/institution ranges or controlled codes.
3	AAAAAAAAA	Registry server	Issue unique participant values within the allocated range.
4	XX	Registry server	Calculate deterministic integrity check digits.

AUTHORITY INVARIANT

KK and NNN are assigned by Osnias. AAAAAAAAAA is assigned by the registry server. XX is calculated by the registry server. No field may be silently reassigned by a wallet, user interface or smart contract.

13. API and Resolution Model**13.1 Lookup request**

```
GET /api/osnias-id/01-001-000000001-37
```

13.2 Example response

```
{
  "osnias_id": "01-001-000000001-37",
  "status": "ACTIVE",
  "checksum_valid": true,
  "participant_type": "ENTITY",
  "wallets": [],
  "external_identifiers": {}
}
```

The example uses a real checksum value. Wallet bindings and optional external identifiers are intentionally omitted from this public example rather than represented by placeholder values.

14. Normative Checksum Specification

Version 1.0 defines the two-digit field XX deterministically. The checksum uses decimal MOD-97 arithmetic over the 14-digit payload KK NNN AAAAAAAAAA

14.1 Input

Remove display separators. Let B be the 14-digit decimal string:

```
B = KK || NNN || AAAAAAAAAA
```

14.2 Calculation

1. Append two zero digits to B, producing B00.

2. Interpret B00 as a non-negative decimal integer.
3. Compute $r = B00 \bmod 97$.
4. Compute $XX = 98 - r$.
5. Encode XX as exactly two decimal digits, including a leading zero if required.

14.3 Validation

Let I be the complete 16-digit identifier B || XX. A structurally valid checksum satisfies:

$$I \bmod 97 = 1$$

STANDARDIZATION NOTICE

This is the normative Osnias-ID checksum rule. It uses MOD-97 arithmetic and is designed to be independently reproducible. This document does not claim formal ISO/IEC 7064 conformity or certification.

15. Checksum Test Vectors

KK	NNN	AAAAAAAA	XX	Canonical display	Validation
01	001	000000001	37	01-001-000000001-37	full numeric ID $\bmod 97 = 1$
01	001	000000002	34	01-001-000000002-34	full numeric ID $\bmod 97 = 1$
12	034	012345678	59	12-034-012345678-59	full numeric ID $\bmod 97 = 1$
99	099	999999999	77	99-099-999999999-77	full numeric ID $\bmod 97 = 1$
05	007	123456789	69	05-007-123456789-69	full numeric ID $\bmod 97 = 1$

15.1 Worked example

For Osnias-ID 01-001-000000001-37:

```

B      = 010010000000001
B00    = 01001000000000100
r      = B00 mod 97 = 61
XX     = 98 - 61 = 37
I      = 01001000000000137
I mod 97 = 1  -> VALID

```

15.2 Error detection

The checksum is intended to detect common transcription and data-entry errors. It is not a cryptographic signature, MAC, authentication token or proof of registry membership. Registry status must still be checked.

16. Formatting Rules

- Canonical display: KK-NNN-AAAAAAAAA-XX.
- Canonical machine value: 16 decimal digits without separators.
- Leading zeroes are significant and must be preserved.
- Spaces are not permitted in the canonical value.
- The checksum is calculated from the 14 payload digits only.
- Formatting separators are excluded from checksum calculation.

17. Uniqueness and Re-Issuance

The registry must not silently recycle an Osnias-ID previously assigned to a different participant. If an identifier is revoked, the historical record should remain reserved or tombstoned to prevent identity ambiguity.

NON-REUSE PRINCIPLE

A previously issued Osnias-ID should remain historically unique even after participant closure, revocation or migration.

18. Lifecycle States

State	Meaning	Clearing admission
PENDING	Record created but not fully activated.	Rejected
ACTIVE	Participant and wallet binding valid.	Eligible, subject to other rules
SUSPENDED	Temporarily disabled.	Rejected
REVOKED	Permanently withdrawn.	Rejected
ARCHIVED	Historical record retained for audit.	Rejected

19. Security Model

- Fail closed when registry resolution is unavailable or ambiguous.
- Separate registry-administration credentials from ordinary user credentials.
- Log all identifier creation, wallet binding, replacement, suspension and revocation actions.
- Prevent client-side generation of authoritative Osnias-ID values.
- Validate checksum before database lookup to reject malformed identifiers efficiently.

- Apply authorization independently of checksum validity: a correct checksum does not imply an ACTIVE record.
- Protect private KYC/KYB data independently from public identifier resolution.

20. Smart-Contract Boundary

Where Osnias-ID is referenced by a smart contract, the contract should treat it as an opaque participant reference or hashed/encoded representation according to the implementation requirements. Smart contracts should not become the authoritative repository of private identity data.

The authoritative legal/KYC mapping remains in the controlled registry layer. On-chain data should be limited to what is necessary for authorization, audit or deterministic linkage.

21. Payment Message Integration

In Osnias Clearing, a payment request must include an Osnias-ID of the payer and/or payee in addition to the execution wallet address. The receiving interface can resolve the participant before acceptance, reducing reliance on raw hexadecimal addresses.

- The Osnias-ID identifies the clearing participant.
- The wallet address identifies the signing or receiving endpoint.
- The payment request identifies the business instruction.
- The settlement transaction identifies the value-transfer execution.

MESSAGE-LAYER PRINCIPLE

Identity, instruction and settlement are separate records. Linking them improves auditability without collapsing them into a single opaque transaction.

22. Institutional Interoperability

For banks, escrow providers, lenders or regulated partners, Osnias-ID can operate as the internal clearing-network participant key used to reconcile technical addresses with institutionally recognized counterparties.

This architecture is particularly relevant where a partner requires KYC/KYB, AML controls, whitelisting, jurisdictional restrictions or a controlled participant directory before permitting a clearing operation.

23. Privacy and Data Protection

- Apply data minimization: store only fields required for the clearing purpose.
- Separate public identifier data from confidential identity evidence.
- Define retention and deletion rules for underlying personal data according to applicable law.
- Preserve audit evidence where legally or operationally necessary without exposing unnecessary personal data.
- Avoid placing personal addresses, e-mail addresses, telephone numbers or identity documents directly on-chain.
- Use access controls and purpose limitation for registry queries containing non-public data.

Osnias-ID itself is an identifier. Whether it constitutes personal data in a given deployment depends on whether it can be linked, directly or indirectly, to an identifiable natural person and on the applicable jurisdiction.

24. Auditability and Governance

The identifier system should be auditable at both the structural and registry levels.

- Structural audit: format, field lengths, checksum and chain/network code allocation rules.
- Registry audit: allocation authority, uniqueness, lifecycle and wallet binding history.

- Security audit: privileged administration, API authorization and logging.
- Privacy audit: data minimization, access control and retention.
- Protocol audit: enforcement of 'no Osnias-ID, no clearing request' at the relevant admission points.

Changes to KK or NNN allocation policy should be versioned and published. The checksum rule should not be changed without a specification-version transition and migration plan.

Annex A — Machine Validation Pseudocode

```
function validateOsniasId(input):
    digits = removeHyphens(input)

    if length(digits) != 16:
        return false

    if digits contains non-decimal characters:
        return false

    payload = digits[0:14]
    receivedXX = digits[14:16]

    remainder = integer(payload + "00") mod 97
    expectedXX = 98 - remainder
    expectedXX2 = leftPad2(expectedXX)

    if receivedXX != expectedXX2:
        return false

    return integer(digits) mod 97 == 1
```

A production implementation must additionally resolve the identifier against the authoritative registry and verify lifecycle status. Structural validation alone is insufficient.

Annex B — KK Chain / Network Code Allocation Rules

This annex uses KK consistently as the Osnias-assigned blockchain/network code. Earlier drafting references to an 'RR namespace', 'RR allocation' or 'RR code' are superseded and should be read as KK. The v0.6 semantic meaning is retained: KK identifies the chain or network allocation used by Osnias Clearing.

- KK contains exactly two decimal digits. KK=00 is permanently reserved as a null/unassigned network code. KK=01–99 therefore provide 99 allocatable namespace positions, subject to reserved values and Osnias allocation policy.
- KK is allocated and controlled by Osnias Clearing as the code identifying a supported blockchain/network domain.
- A KK value identifies a blockchain or network supported by the Osnias Clearing architecture. The detailed chain registry (name, native chain identifier, environment, escrow capability and status) is maintained separately from the 16-digit Osnias-ID so that technical metadata can evolve without changing the identifier format.
- Chain/network allocation policy is external to the checksum and may evolve without changing the checksum formula.
- Reserved KK values must not be used for participant identifiers until assigned to a supported chain/network by Osnias Clearing.
- Deprecated or retired KK chain/network codes remain historically documented and must not be silently reassigned.

TERMINOLOGY CORRECTION

The canonical field name is KK. RR is not part of the v1.0 Osnias-ID format.

Annex C — Reference Test Vectors

01-001-000000001-37	VALID
01-001-000000002-34	VALID
12-034-012345678-59	VALID
99-099-999999999-77	VALID
05-007-123456789-69	VALID

Implementations should include these vectors in automated tests. A single digit change should normally invalidate the checksum.

Annex D — Standards and External References

The following references are provided for interoperability context only:

- ISO 20022-1:2026 — Financial services — Universal financial industry message scheme — Part 1: Metamodel.
- ISO 20022-3:2026 — Financial services — Universal financial industry message scheme — Part 3: Modelling.
- ISO/IEC 7064:2003 — Information technology — Security techniques — Check character systems.
- ISO 17442 series / Global Legal Entity Identifier (LEI) system — external legal-entity identification framework.

Osnias-ID remains a proprietary project identifier unless and until an external standards body, regulator or other competent authority explicitly determines otherwise.

Revision Note — Release 1.0

- Released as Version 1.0 on 9 September 2026.
- Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).
- Reserved KK=00 permanently as the null/unassigned network code, leaving KK=01–99 as 99 allocatable namespace positions.
- Restored Section 8 on multi-address participant bindings and corrected publication-level English and numbering.
- Added canonical author metadata on page 0: Denis Bouzon, ORCID 0009-0007-8894-8902, Osnias Clearing.
- Corrected RR terminology to KK and preserved the v0.6 semantics of KK as the blockchain/network code; removed the draft v0.7 wording that broadened KK into policy or jurisdictional namespaces.
- Defined a deterministic two-digit MOD-97 checksum rule.
- Added five official checksum test vectors and a worked example.
- Replaced the API placeholder XX with a calculated value.
- Clarified that MOD-97 use does not constitute an ISO/IEC 7064 conformity claim.
- Retained explicit no-certification / no-endorsement language for ISO, LEI, UID and FINMA references.
- Corrected the §13.2 calculation steps to an explicit local 1–5 sequence for stable rendering across Word and PDF conversion.
- Removed residual API placeholders: public examples now omit undisclosed wallet and external-identifier values instead of displaying ellipses or null placeholders.

END OF DOCUMENT